

ANALYSTLAB AFRICA

Nigeria Health and Economic Development Analysis

World Bank World Development Indicators (WDI), 2010–2022

A data analytics capstone report examining life expectancy, health expenditure, GDP per capita and population trends in Nigeria.

AnalystLab Africa Data Analytics Capstone
Power BI Dashboard Project

September 2026

1. Project Objective / Problem Statement

This project analyzes selected health and development indicators for Nigeria from 2010 to 2022. The analysis focuses on life expectancy at birth, current health expenditure as a percentage of GDP, GDP per capita and total population. The objective is to identify important trends and relationships that can support evidence-based health and economic decision-making.

Problem statement: Nigeria has experienced substantial demographic and economic changes while improvements in population health have been gradual. A clear data-driven view of how health and economic indicators changed together can help stakeholders identify areas requiring sustained investment and policy attention.

2. Dataset Description

The project uses the World Bank World Development Indicators (WDI). A focused Nigeria working dataset was prepared for Power BI covering 2010–2022 and four indicators.

Indicator	WDI code	Unit
Life expectancy at birth, total	SP.DYN.LE00.IN	Years
Current health expenditure	SH.XPD.CHEX.GD.ZS	% of GDP
GDP per capita	NY.GDP.PCAP.CD	Current US\$
Population, total	SP.POP.TOTL	Persons

The final Power BI-ready table contains 13 annual observations and five fields: Year, Life Expectancy, Health Expenditure, GDP per Capita and Population.

3. Data Cleaning and Preparation

The original WDI bulk file contains many countries, indicators and year columns. For this capstone, the data was narrowed to Nigeria and the indicators relevant to the selected health-and-development theme. The final table was standardized so it could be loaded directly into Power BI.

- Selected Nigeria as the country of analysis.
- Selected four relevant WDI indicators.
- Limited the analysis to 2010–2022.
- Standardized column names and retained one row per year.
- Checked numerical fields for Power BI aggregation and visualization.
- Preserved the units of measurement for correct interpretation.

Note: The Power BI table is a focused analytical extract from WDI rather than the complete WDI bulk-download file.

4. Analysis Methodology

The workflow followed the capstone process of defining the problem, preparing and cleaning the data, analyzing trends, visualizing results in Power BI, interpreting findings and developing recommendations.

- Descriptive statistics: averages, minimums, maximums and changes.
- Time-series analysis: annual comparisons from 2010–2022.
- Trend visualization using line charts.
- Comparative visualization using a combined column-and-line chart.
- Descriptive correlation analysis; correlation does not establish causation.

5. Key Findings

Life expectancy increased: from 51.315 years in 2010 to 54.079 years in 2022, an increase of 2.764 years (about 5.4%).

Health expenditure increased overall: from 3.30% of GDP in 2010 to 4.21% in 2022, a rise of 0.91 percentage points, with fluctuations during the period.

GDP per capita was volatile: it ranged from about US\$1,876.34 in 2017 to US\$3,189.81 in 2019.

Population grew substantially: from 166,642,886 in 2010 to 223,150,896 in 2022, approximately 33.9% growth.

Descriptive relationships: life expectancy correlated at about 0.327 with GDP per capita and 0.524 with health expenditure. These are associations, not causal effects.

Indicator	2010	2022	Change
Life expectancy (years)	51.31	54.08	+2.76 years
Health expenditure (% GDP)	3.30%	4.21%	+0.91 pp
GDP per capita (US\$)	\$2,202.26	\$2,899.16	+31.6%
Population	166,642,886	223,150,896	+33.9%

6. Dashboard Design and Visualizations

The Power BI dashboard provides a simple interactive view of Nigeria's health and development trends. It contains four KPI cards, four individual trend charts, one combined comparison chart and a year slicer.

- KPI cards: Average Life Expectancy, Average Health Expenditure, Average GDP per Capita and Average Population.
- Life Expectancy Trend, Health Expenditure Trend, GDP per Capita Trend and Population Trend.
- GDP per Capita and Life Expectancy comparison using a combined column-and-line chart.
- Year slicer for interactive filtering.

7. Insights

Nigeria recorded a gradual improvement in life expectancy during the study period, while population growth was much faster. This creates continuing pressure on health and social systems.

Health expenditure increased overall but was not stable. Sustained and predictable investment is therefore important rather than relying only on short-term increases.

GDP per capita showed considerable volatility, demonstrating that economic conditions should be considered alongside health outcomes when planning development interventions.

The descriptive relationships suggest that life expectancy moved moderately with health expenditure and more weakly with GDP per capita in this focused extract. Other factors such as education, healthcare access, disease burden, sanitation and inequality were not included and may also influence outcomes.

8. Recommendations

1. **Sustain health-sector investment:** Maintain predictable funding for primary healthcare, preventive services and essential health programmes.
2. **Strengthen primary healthcare access:** Prioritize accessible services for underserved populations.
3. **Link health planning to population growth:** Use demographic trends when planning facilities, health workers, medicines and infrastructure.
4. **Protect essential health spending during economic volatility:** Develop financing strategies that preserve priority programmes when economic conditions weaken.
5. **Monitor outcomes as well as spending:** Track life expectancy and other health outcomes alongside expenditure.
6. **Expand future analysis:** Include maternal mortality, infant mortality, immunization, poverty, education and healthcare access for a broader development assessment.

9. Conclusion

This project demonstrates how World Bank development data can be transformed into an accessible decision-support dashboard. From 2010 to 2022, Nigeria recorded gradual improvement in life expectancy, overall growth in health expenditure, substantial population growth and volatile GDP per capita. The findings emphasize sustained health investment, population-aware planning and resilience during economic fluctuations.

10. References

World Bank. (2026). *World Development Indicators (WDI)*. World Bank Data Catalog.

World Bank. *Life expectancy at birth, total (years)* — SP.DYN.LE00.IN.

World Bank. *Current health expenditure (% of GDP)* — SH.XPD.CHEX.GD.ZS.

World Bank. *GDP per capita (current US\$)* — NY.GDP.PCAP.CD.

World Bank. *Population, total* — SP.POP.TOTL.